Spatial Distribution of Bidirectional Charging Stations by Simulation and Evaluation of Traffic and Grid Data

Achim Baumgärtner[1], Erik Weigand[1],
Gian Bruno Truöl[1,2], Mirko Sonntag[1,2]

[1]University of Applied Sciences Esslingen, Esslingen University, Germany
[2]Fraunhofer Application Center Keim, Esslingen, Germany

acbait01@hs-esslingen.de, erwewi01@hs-esslingen.de,
gian.ledl-truoel@hs-esslingen.de, mirko.sonntag@hs-esslingen.de

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Abstract. The goal of this paper is to investigate how bidirectional charging stations can be distributed across a dynamic area by combining traffic simulation with partially extracted grid data from official sources. We model charging demand as a spatiotemporal process and evaluate candidate locations with respect to accessibility and traffic flow. The proposed workflow integrates mobility traces, station utilization diagnostics, and synthetic grid constraints as well as a deployment strategy for simplified setup and use. Results indicate that jointly considering traffic and grid criteria improves charging availability in high-demand districts and helps identify shortfalls in electrical grid capacity. Across five scenario instances the fleet's end-of-day state of charge rises by about three percentage points and the share of undersupplied vehicles falls from 44 to 37 percent, with the number of stations mattering more than the strategy used to place them.

Keywords: Bidirectional charging, traffic simulation, SUMO, OpenStreetMap, spatio-temporal analysis, charging demand, V2G

Nomenclature

AC	Alternating Current
BEAM	Behavior, Energy, Autonomy, and Mobility
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
EV	Electric Vehicle
MATSim	Multi-Agent Transport Simulation
OSM	OpenStreetMap
POI	Point of Interest
SoC	State of Charge
SUMO	Simulation of Urban Mobility
TraCI	Traffic Control Interface
V2G	Vehicle-to-Grid

1. Introduction

The rapid growth of electric mobility is shifting charging-infrastructure planning from static placement rules toward data-driven decision support [1]. Bidirectional charging adds a further dimension, since vehicles are no longer only energy consumers but can also feed power back to the grid, a concept known as Vehicle-to-Grid (V2G), which turns charging points into distributed flexibility assets [2]. Deciding where to place such stations therefore requires understanding where and when charging demand actually concentrates.

Because electric vehicles still represent a small share of the fleet, there is little real-world data on where charging stations should be placed and how intensively they are used. At the same time, uncoordinated charging concentrates electrical load in space and time and can stress local distribution grids [3], which makes the spatial and temporal structure of demand a central planning question. Existing siting studies, such as [4], however, often rely on data or tooling that is not openly reproducible. We therefore ask: *how can candidate locations for charging stations be ranked from openly available data, based on their accessibility and on how charging demand is distributed over space and time?*

To address this, we build a simulation workflow on openly available data. Road-network and land-use information are taken from OpenStreetMap (OSM), and traffic is simulated microscopically in Simulation of Urban Mobility (SUMO), controlled through its Python interface Traffic Control Interface (TraCI). From POIs we derive travel demand, and model charging as a *spatiotemporal* process, that is, demand that varies simultaneously by location and time of day. The model distinguishes public charging stations from private home wallboxes with vehicle-specific access control. A custom web interface, which is based on SUMO's web interface, lets users select an area of interest on a map and generate ready-to-run scenarios directly, lowering the barrier to reproducing and extending the analysis. Candidate sites are then assessed in terms of accessibility, station utilization, and the resulting local charging load.

The scope of this study is the planning and evaluation stage. We focus on the spatial and temporal distribution of charging demand under realistic mobility patterns. The distribution grid is represented by a synthetic network whose power flow bounds the power available at each station, so that the charging we report stays within its voltage and loading limits. Analysing that grid is not the subject of this study. Peak load, hosting capacity and reinforcement planning are left to future work, as are battery degradation and market or pricing mechanisms.

1.1 Motivation

Urban charging demand is highly uneven across time and location, which might cause queues at highly frequented stations while other sites remain virtually unused. At the same time, cars that are parked for extended periods of time (in residential or commercial areas) can remain plugged in, creating opportunities for bidirectional energy exchange. The spatial distribution of charging stations therefore affects both user experience and grid load. Understanding the spatial and temporal patterns of charging demand is a prerequisite for infrastructure planning. However, there is a lack of open, reproducible methods for assessing candidate locations based on real-world mobility patterns and grid constraints.

1.2 Problem Statement

Candidate locations for bidirectional charging stations need to be evaluated with respect to their accessibility and traffic, while taking into account the constraints of the electrical grid. To solve this problem, a workflow is proposed that integrates a demand model that resolves where and when charging demand occurs, a placement rule that turns the demand into candidate stations, and an evaluation that separates the contribution of the placement rule from that of simply adding capacity.

1.3 Contributions

The research contributes a practical and reproducible workflow for planning bidirectional charging infrastructure from openly available data. This work contains a user interface for scenario-based traffic generation along with an evaluation of candidate station portfolios under varying demand conditions, which allows competing placement strategies to be compared on the same footing rather than described in isolation.

The main contributions are as follows:

- A simulation-based siting pipeline that derives charging demand from land-use data and evaluates candidate stations against the mobility patterns that produced it.
- An empirical comparison of two placement strategies across repeated scenario instances, which quantifies how much of the accessibility gain is attributable to the number of stations rather than to where a heuristic places them.

- A reference implementation in Python to support reproducible experiments and scenario-based sensitivity studies.

2. Related Work

Planning charging infrastructure has been studied extensively, and recent reviews organize the field along the dimensions of station siting, sizing, grid reinforcement, trip-considerations and the treatment of uncertainty [5, 13]. Most contributions optimize either where demand is best served or how the grid copes with the additional load [14–16, 18]. Bidirectional charging, in which vehicles also feed power back, is only beginning to be integrated into such planning frameworks [9].

2.1 Traffic-Aware Siting

Traffic-aware siting methods estimate charging demand from origin-destination flows, activity patterns, and route choices, often placing stations so as to maximize coverage of traffic along routes. Data-driven analyses suggest that charging demand is largely shaped by proximity to amenities, Electric Vehicle (EV) density, and adjacency to high-traffic corridors, which motivates deriving demand from land-use data as we do [6]. Microscopic traffic simulators are increasingly used for this purpose. SUMO in particular provides built-in energy and charging-infrastructure models and has been used to assign electric vehicles to charging stations from simulated traffic data [7, 8].

A different approach is agent-based traffic simulation, which represents the road network as nodes and links and derives traffic flow from travel demand and road capacity. This approach can be used to estimate charging demand at a more aggregated level, and can be useful for planning charging infrastructure in areas with limited data availability or where detailed traffic simulation is not feasible. Multi-Agent Transport Simulation (MATSim) is an example of such a model that has been used for this purpose, and paired with Behavior, Energy, Autonomy, and Mobility (BEAM) it is possible to simulate the impact of charging infrastructure on a dynamic traffic network [11, 12]. A related family of methods derives candidate sites by clustering demand instead of optimising over a fixed candidate set, using dwell events extracted from taxi trajectories or recorded charging sessions [19, 20]. The signal we cluster is different. It is the position at which a simulated vehicle falls below 30% State of Charge (SoC), which marks where energy is actually needed rather than where vehicles happen to stand.

2.2 Coupled and Bidirectional Approaches

A smaller but growing body of work couples transportation simulation with power-system analysis, for example through co-simulation frameworks that pass simulated charging loads to a distribution-grid model [10]. For bidirectional charging, planning models increasingly distinguish public and private chargers and account for uncertainty in demand and renewable generation [9]. Integrated and openly reproducible frameworks that combine microscopic mobility simulation with bidirectional charging at the level of individual public stations and private home wallboxes, however, remain limited. This paper addresses that gap on the mobility-and-demand side. The distribution grid enters as a synthetic network that constrains the power available at each station, while its analysis is left to future work.

3. Methodology

SUMO is used as the microscopic traffic simulator and TraCI for real-time control, combined with a grid-impact model and a statistical distribution of home-charging devices, to evaluate the spatial distribution of bidirectional charging stations. Two siting strategies (*Saturate-and-Prune* and *Clustering*) are implemented and described in detail in section 3.2.

3.1 System Overview

Figure 1 illustrates the complete workflow pipeline. The framework is organized into four

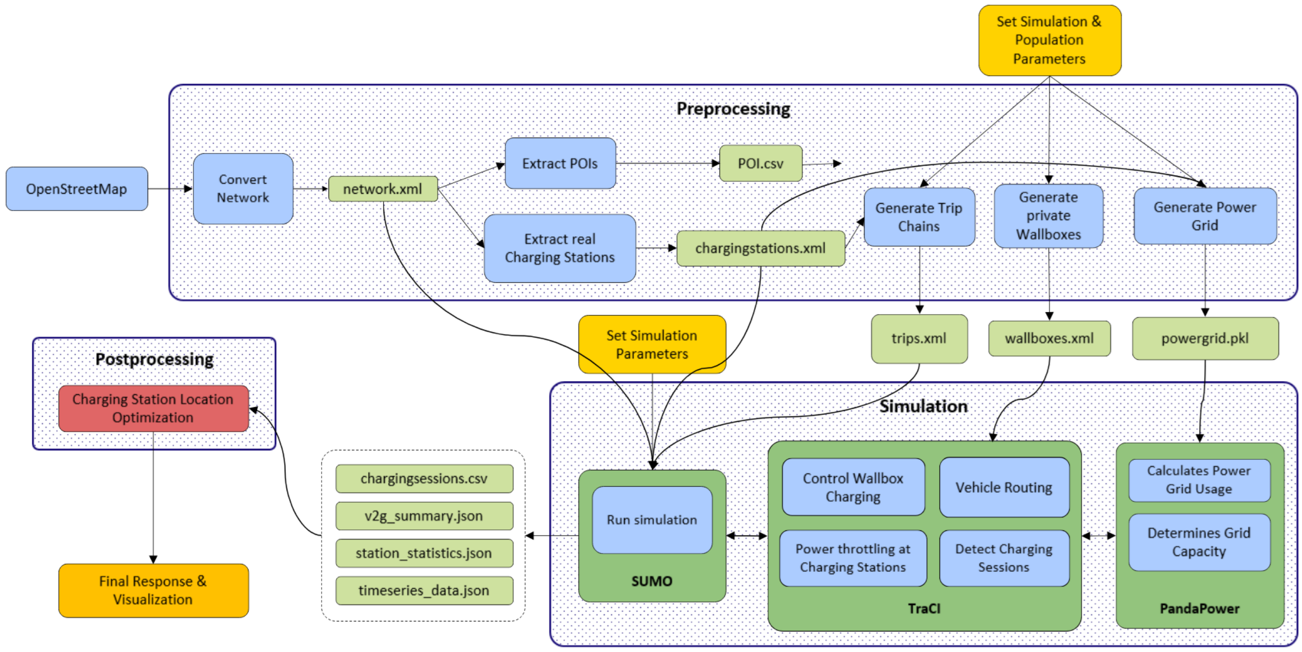

Figure 1: Complete Workflow Pipeline: From OSM data acquisition to scenario generation, execution and manipulation to analysis and visualization.

consecutive stages: (i) data acquisition through OpenStreetMap, (ii) preprocessing and scenario generation, (iii) coupled traffic–grid simulation, and (iv) postprocessing, analysis and visualization. Each stage produces intermediate artifacts that are consumed by the subsequent stage, so that the entire process is reproducible from a single set of user-defined parameters.

Data acquisition. The pipeline retrieves information like the road network, electrical grid topology, and candidate Point of Interests (POIs), which are extracted from OSM. The road network is converted into a SUMO-compatible format using the `netconvert` tool by SUMO, while grid features and POIs are persisted for later preprocessing.

Preprocessing and scenario generation. The `chargingstations.xml` file contains both existing charging stations from OSM and candidate stations produced by the placement algorithm. The file is utilized not only for the generation of the synthetic power grid, but also in assistance with the POI file to generate trip chains. Trip chains map individual agents to a sequence of activities and destinations, from which their charging demand and low-SoC behavior emerge. In parallel, the private-wallbox generator assigns home-charging devices to a share of agents following a statistical distribution, and the power-grid generator couples the charging stations and wallboxes to their nearest buses to form the synthetic distribution grid. To finalize the preprocessing, user-set parameters (EV penetration rate, wallbox share, battery capacities, grid voltage levels) are applied to the mechanisms of the trip chain, private wallbox, and power grid generation, yielding a fully specified scenario. Scenario generation is seeded and therefore deterministic. The coupled simulation is deterministic only once the interpreter's hash seed is fixed as well, as will be discussed in section 4.

Coupled traffic-grid simulation. SUMO reads `private_wallboxes.xml`, `osm.passenger.trips.xml` and `power_grid-.pkl` and simulates traffic and charging in time steps of fixed length. SUMO and PandaPower never exchange data directly. TraCI sits between them over a Python interface, passing charging loads to the grid model and returning the power each station may draw, which is

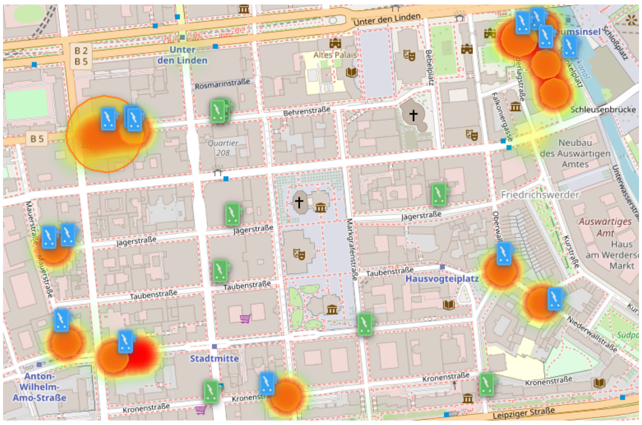

Figure 2: Resulting visualization with generated (blue) and extracted (green) CS for example run.

what lets grid limits act on vehicle behaviour.

Postprocessing, analysis and visualization. The simulation produces a variety of output files, including `station_statistics.json`, `v2g_summary.json`, `charts_data.json`, `charging_sessions.csv` and `model_log-_data.csv`. These persisted outputs are analyzed and visualized to evaluate the placement algorithm along metrics such as charging sessions, net energy flow over time, peak grid power, and individual charging station utilization (section 3.3). The results are rendered as plots and maps in the web interface, and inform future iterations of the placement algorithm. Figure 2 shows an example visualization with generated (blue) and extracted (green) charging stations for a single run. The heatmap and orange circles indicate approximate areas for charging station placement, while the blue and green dots represent the actual locations of the generated and extracted charging stations, respectively.

3.2 Charging Station Placement Algorithm

Two siting strategies are implemented to compare different placement policies:

- **Saturate-and-Prune**: A deliberately naive iterative approach. Candidate stations are first placed on every road segment long enough to hold one, which guarantees maximum coverage but far exceeds any plausible build-out. The scenario is then simulated and the least utilised candidates are removed before the next iteration, where utilisation is the fraction of simulation steps during which a station was occupied and the threshold is 5 %. Saturating the network dilutes utilisation. The same demand spread over many more stations leaves almost none of them above any fixed threshold, so removing every station below it in one pass empties the candidate set immediately. At most half of the remaining candidates are therefore removed per iteration, least utilised first, which lets demand re-concentrate on the survivors between iterations. Stations that already exist in the area are exempt from pruning, because existing infrastructure cannot be removed in practice and the strategy is meant to decide where to add capacity. The loop is run for a fixed number of iterations rather than to convergence, since each iteration requires a full simulation.
- **Clustering with DBSCAN**: A heuristic that prioritizes locations with high traffic flow, potential traffic congestion, and consideration for vehicles that enter a state of low SoC. Placement follows the traffic flow, aiming to maximize accessibility where vehicles are most likely to need charging while reducing detours. DBSCAN or Density-Based Spatial Clustering of Applications with Noise, specifically, is a clustering algorithm that groups together close points based on a distance metric and a minimum number of points, while marking points that are too far away as noise. DBSCAN has the advantage of not requiring a predefined number of clusters or an arbitrary shape.

3.3 Evaluation Metrics

Mobility accessibility. Placement is evaluated from the charging behaviour and the state of charge that the simulated fleet exhibits over a

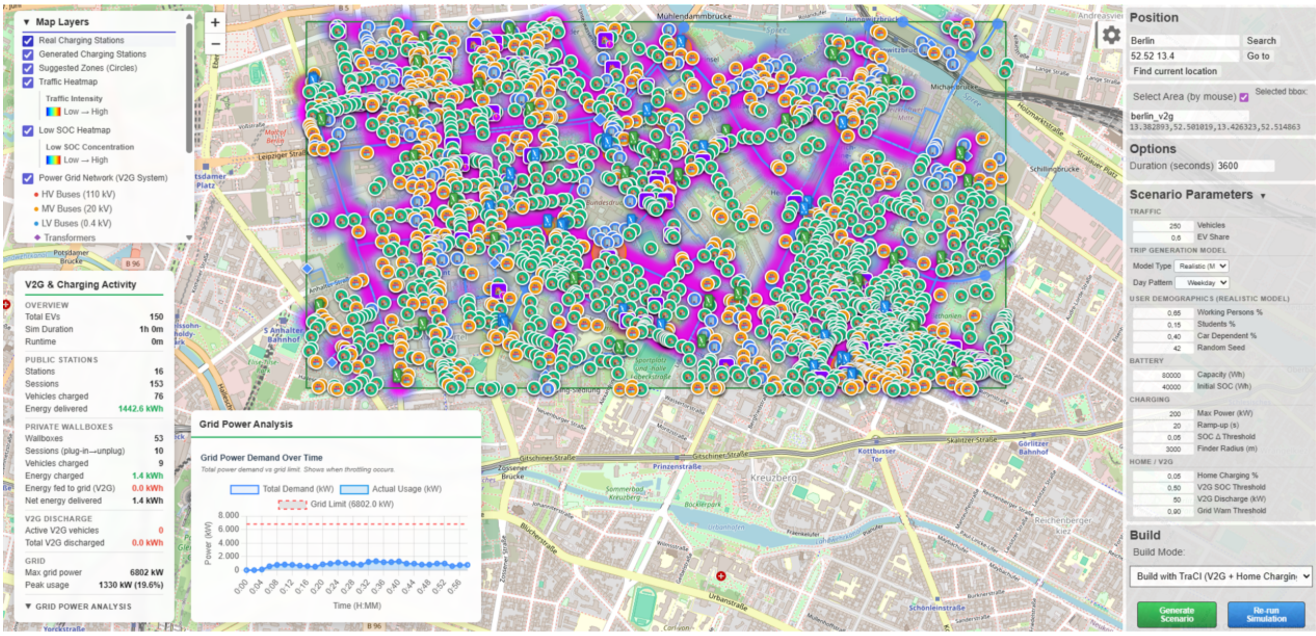

Figure 3: Example run of the workflow: the web interface with all data layers enabled.

full day. We report the share of electric vehicles that charge at a public station at least once, the number of stations that deliver energy at all, and the distribution of the state of charge at the end of the simulated day, summarised by its mean and median. Because the mean alone hides how the burden is distributed, we additionally report the share of agents ending below 20% SoC, the share dropping below 10% at any point during the day, and the share that is fully depleted. Together these describe whether the network serves the fleet broadly or concentrates service on the vehicles that happen to pass a reachable station.

Grid impact. During the simulation, PandaPower solves an Alternating Current (AC) power flow on the synthetic distribution grid at every step, evaluating bus voltages and the loading of lines and transformers as constraints that bound the power allocated to each station. Grid capacity additionally constrains the sizing of candidate sites. Of 147 candidates generated across the five scenario seeds, 41 were limited in their charge-point count by the capacity available at the nearest connection point. The grid therefore acts as a constraint on the placement study rather than as an object of it. We do not report grid quantities such as peak load, and we make no claim about hosting capacity or reinforcement need.

4. Experiments

4.1 Datasets

The experiment is conducted on mobility data, which is generated by an algorithm that simulates movement patterns of agents in a given area. The algorithm takes into account various factors such as land use, transportation infrastructure and natural behavior of agents to generate realistic mobility data. The generated data is then used to evaluate the performance of the proposed charging station placement algorithm. Before running the simulation, the dataset consists of surface level mobility (agent start- and endpoints, OSM-data), which is used for traffic flow, POIs and electrical grid information. When a simulation is run, the dataset is enriched with additional information such as charging behavior, energy consumption, specific locations of agents and their SoC. The datasets are generated using a combination of real-world data and synthetic data. Whether results obtained this way transfer to other districts or fleet sizes is not tested here. The datasets are also designed to be scalable, allowing for the evaluation of the placement algorithms on larger areas with more agents and charging stations.

4.2 Experimental Setup

The reference scenario covers a one kilometer by one kilometer area in Berlin, Germany. Each agent follows a daily trip chain with departures spread across the day and returns to its home location between activities. The simulation covers 24 hours of simulated time with a time step of 10s, so the chains complete within the horizon and the evening period at home is part of the observation. The trip generator is selected by the pipeline from the configured duration. Horizons below ten hours use a compressed departure window intended for short test runs, longer horizons use the daily generator applied here.

Every experiment is repeated over five random seeds, which redraw home and activity locations, the assignment of electric vehicles, and departure times. Across those five populations the scenario holds 236.6 $\pm$ 3.4 agents, of which 142.6 $\pm$ 1.8 are electric vehicles with a battery capacity of 80kWh initialised at 50% SoC, the remainder are conventional vehicles. Every agent carries a SUMO battery device, because the simulation is configured with `device.battery.probability` = 1, so the fleet-wide SoC statistics cover all agents that entered the network, while charging is restricted to the electric vehicles. *690,000 private charging points* were subsidised in Germany in 2021 and 2022, but half of the electric vehicle owners are assigned a private wallbox at their home lane [17]. The aim is to assess how many agents would use their respective wallboxes when given the opportunity. The wallboxes are deliberately excluded from the additional file handed to SUMO and exist only inside TraCI, which restricts each of them to its owning vehicle type. As a result SUMO's stationfinder cannot route any vehicle to a private wallbox, and home charging occurs only when an agent is parked at its own home location.

Charging power, V2G discharge and the wallbox logic are injected at runtime through TraCI, which overrides the station power, applies the ramp-up, and arbitrates the total power draw against a synthetic distribution grid modelled in PandaPower with 37 buses, 93 lines and 10 transformers. Charging power per station is capped at 200kW with a 20s ramp-up, V2G discharge is enabled above 50% SoC with up to 50kW per vehicle and is triggered when the grid reaches 90% of its capacity. The grid capacity is derived from the area of the scenario at a density of 1500kW per square kilometer, giving 1500kW here.

The public infrastructure of the reference case is the one already present in the area, that is, the 26 charging stations extracted from OSM. Each station is projected onto the nearest road segment whose lane is long enough to hold it, which keeps all 26 stations in the network. Otherwise, placing them on the closest segment regardless of their length would put several of them on junction stubs of a few decimetres, where the resulting extent would be degenerate and rejected.

Reproducibility. Repeated executions of an identical configuration were initially found to differ, with a spread of roughly 7% in public and 9% in private charging energy. The cause is the iteration order over the set of active electric vehicles. It determines the order in which vehicles request power from the shared pool and which varies between processes because Python randomises string hashing. Fixing the interpreter hash seed makes a run bit-identical, so all runs reported here are executed with a fixed hash seed and the variation between them is due to the scenario seed alone.

4.3 Results

4.3.1 Scenario generation and simulation

Every scenario begins with a bounding box drawn on a map. OSM supplies the road network, the land-use POIs, the charging stations already installed and the features behind the synthetic grid, and no second data source is involved. From a single parameter set the pipeline then derives travel demand, assigns private wallboxes to their owners and writes the station and grid files. Choosing an area and pressing one button is enough to generate a scenario and run it, since every parameter the interface exposes carries a default.

This study produced 25 scenarios and 50 coupled simulations. A full simulated day at a 10s step took 189s on average, which puts the whole study at 2.6 hours of compute on one desktop machine. That cost is what made the control arms in section 4.3.4 affordable.

4.3.2 The existing infrastructure

Where the energy is drawn. Over the simulated day the fleet draws 3853kWh, of which public stations deliver 3251.7 ± 315.4kWh in 246.2 ± 9.2 charging sessions to 98.2 ± 1.5 vehicles and private wallboxes deliver 601.5 ± 258.2kWh in 27.0 ± 6.5 charging sessions. Public charging thus accounts for 84% of the energy. The imbalance follows from how the two kinds of infrastructure are reachable. Public stations are visible to SUMO's native stationfinder, which enables agents to dynamically change their route, and are approached whenever a vehicle runs low on charge, whereas a wallbox can only be used when its owner happens to be parked at home, which the spread of departure and return times makes comparatively rare within the horizon. The large relative spread of the wallbox figures, 43% of their own mean, reflects exactly that. Whether home charging happens at all depends on which agents a given seed places where. Every wallbox session is attributable to the owning agent, so the type restriction on private stations holds.

Accessibility. Despite the public charging that does take place, the fleet ends the day well below its starting point. The mean SoC drops from 50.0% to 28.0 ± 0.7%, with a median of 33.9 ± 2.9%. Of all agents, 43.9 ± 1.2% end below 20% SoC, 39.0 ± 2.2% drop below 10% at some point during the day, and 3.8 ± 0.3% are fully depleted. Table 1 summarises the reference case.

Stations are not the bottleneck. No queueing occurs at any public station over the simulated day. Of the time vehicles spend connected, 12.9% is spent waiting rather than charging, and that share is accounted for almost entirely by the fixed 200s connection delay. 45 vehicles wait, on average for 3.3 minutes, against a configured delay of 3.3 minutes. No vehicle waits for a station that is occupied by another. The shortfall in table 1 is therefore not a capacity problem at the stations that exist, but a question of which stations a vehicle can reach on the route it drives.

Bidirectional operation. V2G discharge is enabled but contributes no energy in the reference case, and remains negligible in every configuration studied here. Discharge requires a vehicle above 50% SoC, while the fleet ends the day at a mean of 28.0% and the vehicles occupying the stations are there precisely because they are low. The built work therefore demonstrates that the mechanism to charge bidirectionally is in place. To quantify the value of bidirectional charging requires a fleet segment that is parked at a high state of charge when it is needed so that it can contribute to the grid, which is left to future work.

4.3.3 Derived charging demand

The Density-Based Spatial Clustering of Applications with Noise (DBSCAN) stage of the Clustering strategy operates on the samples in which a vehicle is below 30% SoC; half of them are drawn at random to bound the clustering cost. Clusters that fall within 100m of an already existing station are discarded, and across the five seeds the stage retains 29.4 ± 3.8 clusters. These are then turned into 34.6 ± 5.0 charging stations, one per cluster plus one for every further 50 estimated charge points, which is what raises the network from 26 to 60.6 ± 5.0 stations. A sample is one vehicle at one 10s time step, so the sample count measures dwell time rather than demand, and a vehicle parked at home for hours contributes thousands of samples without ever asking for a public charge.

Table 1: Reference case on the existing OSM infrastructure, mean ± standard deviation over five scenario seeds. 24 hours of simulated time, 10s time step, coupled through TraCI.

Metric	Value
Agents	236.6 ± 3.4
of which EV	142.6 ± 1.8
Simulation horizon	24h (8640 steps)
Mean SoC, start → end	50.0 → 28.0 ± 0.7%
Median end SoC	33.9 ± 2.9%
Agents below 20% SoC at end	43.9 ± 1.2%
Agents below 10% SoC (any time)	39.0 ± 2.2%
Agents fully depleted	3.8 ± 0.3%
Public stations available (OSM)	26
Public sessions	246.2 ± 9.2
Public vehicles served	98.2 ± 1.5
Public energy	3251.7 ± 315.4kWh
Wallbox sessions	27.0 ± 6.5
Wallbox energy	601.5 ± 258.2kWh
V2G energy discharged	0.0kWh

The demand of a cluster is therefore derived from the vehicles themselves. Samples inside a cluster are grouped per vehicle and split into separate visits whenever a vehicle is absent for more than five minutes, and each visit is charged with the energy it would need to reach 80% SoC. Table 2 lists the ten strongest clusters of one representative seed. Demand is spread over many sites rather than concentrated in a few, and the mean SoC inside a cluster is well below the 30% threshold, which indicates that the retained clusters mark genuine range emergencies rather than opportunistic stops. Cluster radii stay compact, so each cluster is small enough to be served by a single station site.

4.3.4 Comparison of the placement strategies

Saturate-and-Prune places a candidate on every road segment long enough to hold one, which yields 196 candidates on top of the 26 existing stations, and then removes the least utilised half of the remaining candidates in each iteration. Running the loop for six iterations traces the network down from 222 to 33 stations and yields an accessibility curve against station count rather than a single operating point.

Table 2: Charging-demand clusters with highest energy demand for scenario seed 42. *Vis:* number of distinct vehicle visits, *Veh:* number of vehicles visiting the cluster, *kWh:* energy needed to reach 80% SoC, *CP:* resulting number of charge points and *Cap:* grid capacity at nearest connection point.

ID	Vis. (#)	Veh. (#)	SoC (%)	Rad. (m)	kWh	CP (#)	Cap. (kW)
44	17	14	17.8	47.2	845.5	2	500
35	10	10	17.9	25.0	503.6	1	5000
13	9	9	15.2	32.8	468.8	1	5000
34	7	7	24.9	25.0	352.7	1	5000
45	7	7	22.5	65.3	351.8	1	5000
14	7	7	18.1	45.8	348.5	1	5000
37	5	5	22.6	25.0	250.1	1	5000
53	4	4	17.6	25.0	198.8	1	5000
24	4	4	16.9	48.2	196.7	1	5000
39	4	4	19.0	25.0	193.6	1	5000

All 32 clusters: 111 visits, 5657.4kWh, 33 charge points; SoC 15.6 ± 6.0%; radius 28.7 ± 9.4m; grid rating 15 excellent, 17 good.

Two control trials round out the picture. Both draw 35 sites from the same pool of viable segments and add them to the 26 existing stations, which lands them at 61 stations against the 60.6 that Clustering reaches. One draws at random. The other takes the 35 segments furthest from the demand clusters of that seed, so it is Clustering inverted on identical input. Together these trials show how much of the gain is due to adding stations and how much to choosing them well.

Clustering derives its candidates from the demand log of the reference run in a single pass and ends at 60.6 ± 5.0 stations, which places it between the fourth and third iteration of that curve. Figure 4 shows both.

Count and placement of sites matter. At 61 stations the three selection rules order themselves as designed. Anti-clustering reaches 29.5 ± 1.1% mean end SoC, random placement 30.4 ± 1.6%, and Clustering 31.4 ± 1.0%. Against the 28.0% of the reference case this splits the total gain of 3.4 points almost evenly. Adding

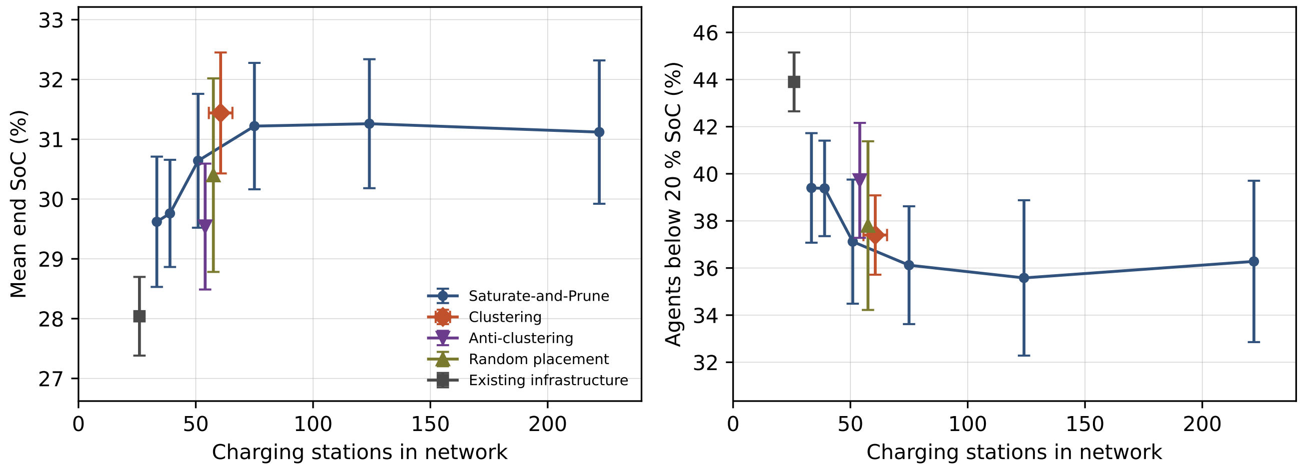

Figure 4: Accessibility against number of charging stations, mean over five scenario seeds with standard deviation as error bars. Left: mean SoC of the fleet at the end of a simulated day. Right: share of agents ending below 20% SoC. The line traces successive prune iterations of Saturate-and-Prune. Clustering produces its network in a single pass. Random placement and anti-clustering are control arms at the same station count, drawn from the same pool of viable road segments. All three arms at that count are nudged apart horizontally so their markers stay readable.

Table 3: The four configurations at a comparable station count, mean ± standard deviation over five scenario seeds. Anti-clustering and random placement draw from the same pool of viable road segments as Saturate-and-Prune, so the three differ only in how the sites are chosen.

	Exist.	Anti	Rand.	Clust.
Stations	26	61	61	60.6
Mean	28.0 ±0.7	29.5 ±1.1	30.4 ±1.6	31.4 ±1.0
Median	33.9 ±2.9	39.9 ±3.9	40.8 ±5.2	46.3 ±1.7
<20	43.9 ±1.2	39.7 ±2.4	37.8 ±3.6	37.4 ±1.7
<10	39.0 ±2.2	34.6 ±2.3	31.3 ±4.2	31.7 ±2.4
kWh	3252 ±315	4454 ±447	4499 ±349	5781 ±530

stations in the worst available positions already recovers 1.5 points. Choosing them well adds the remaining 1.9. Neither placement, nor station count alone decides the outcome. The three methods at 61 stations are summarised in table 3.

Clustering and random placement are not separable on the mean, 1.0 points apart at rather less than one standard deviation. They separate on reliability. Clustering holds a median of 46.3 ± 1.7% where random placement gives 40.8 ± 5.2%, and the same pattern shows in the share of agents below 20% SoC at ±1.7 against ±3.6. A random draw sometimes finds good sites and sometimes does not. The demand-driven rule finds them every time. A network is built once and cannot be averaged over five attempts, which makes the narrower spread the more useful property of the two.

Past 50 stations the mean end SoC sits near 31%, up from 28.0%. The median gains ten points and the share of agents finishing below 20% SoC drops from 43.9% to somewhere between 35 and 37%. The difference in the share of agents finishing below 20% SoC is three to seven times larger than the spread across seeds. Clustering, at its own station count, lands on the Saturate-and-Prune curve. At 60.6 stations, the curve gives about 30.9% mean SoC against the 31.4 ± 1.0% Clustering reaches, half a standard deviation apart. On the two tail metrics the order flips. Clustering leaves 37.4 ± 1.7% of agents below 20% and 31.7 ± 2.4% below 10%, where the curve gives roughly 36.7 and 29.9%. Five seeds cannot tell the two apart. What moves accessibility is not only the number of stations, but also where they are placed.

Coverage saturates early. Between 222 and 75 stations the mean end SoC holds at 31.1 to 31.3%. Stripping stations out even nudges the median upward. Things only start to slip below roughly 50. At 39 stations the mean drops to 29.8 ± 0.9% and agents falling under 10% SoC go from 28.9 to 33.3%. That leaves a working range of about fifty to seventy-five stations for this area. Anything beyond is spare capacity, which explains why 222 stations buy nothing over the 75 left after three prune steps. It also explains the tie between the strategies. Both end up inside the plateau.

Simulation effort. Clustering derives its network from a single simulation of the reference case. Saturate-and-Prune spends one simulation per prune step, six of them here, and where it stops depends on how many steps are run rather than on the 5% utilisation rule, which is rarely triggered. In a planning workflow that matters. On this evidence it is also the only advantage the demand-driven heuristic can claim over naive saturation.

Threshold alone is not a sufficient criterion. One or two of 222 saturated stations clear 5% occupancy, so the criterion marks nearly everything for removal and the half-per-iteration cap sets the actual schedule. Lowering the number would not fix that. Across the resulting networks, 4.8 ± 2.6 stations sit above 5% occupancy, 43.6 ± 1.5 above 1%, and 48.6 ± 0.5 above 0.1%. Occupancy is bimodal. A station is either used or untouched, so anything between 1% and 0.1% produces much the same network. What the implementation really does is halve repeatedly, ranked by utilisation, rather than apply a threshold. To resolve the issue, only a limited amount of stations will be removed per iteration, which lets demand re-concentrate on the survivors between iterations.

5. Discussion

5.1 Practical Example

For the area studied, the curve in fig. 4 shows a range, not a single answer. Between 51 and 75 stations the mean end SoC holds near 31%. Below that it drops, at 39 stations to 29.8%. Above it nothing happens, 222 stations perform no better than 75. For this specific case (a 24-hour simulation in Berlin, with the specific bounding box and fleet) a planner would be advised to add roughly 30 sites to the 26 already existing stations. Those 30 sites move the fleet from a mean of 28.0% to 31.4% and cut the share of vehicles ending below 20% SoC from 43.9 to 37.4%. Both strategies land inside the plateau and neither can be told from the other. A random draw of the same size reaches a similar mean but swings far more between seeds, which matters when a network is built once rather than averaged over five attempts. Saturate-and-Prune is the more expensive of the two in simulation time. Overshooting is the expensive mistake. A network of 222 stations serves the fleet no better than one of 75, at higher cost and complexity. This shows that a plateau exists and can be located for any area with OSM coverage, before a station is built.

5.2 Limitations

Bidirectional operation is barely exercised. V2G discharge is implemented and enabled, but contributes essentially no energy in any configuration studied. Discharge requires a vehicle above 50% SoC, and the vehicles connected to a station are there because they are low, so the flexibility the mechanism needs is not present where it would be used. Quantifying a V2G benefit requires a fleet segment parked at a high state of charge at the relevant time, which points at home charging in the evening rather than at public stations, and is left to future work.

Reachability of private wallboxes. Private wallboxes are restricted to their owning vehicle and are deliberately hidden from SUMO, so that no foreign vehicle can occupy them. The run confirms that the restriction holds, with all 20 wallbox sessions attributable to the owning agent. The same design also means that SUMO's stationfinder never routes a vehicle home to charge. Home charging occurs only when an agent is parked at its own home location for other reasons, which the spread of departure and return times makes rare within a 24-hour horizon. The low private share reported in section 4.3 therefore reflects the opportunity for home charging within this scenario, not a general statement about the role of private infrastructure. Which owners receive one is drawn uniformly at random. Real ownership correlates with building type and parking availability, so an actual district would concentrate home charging in some streets and leave others without it, which would shift the demand map the clustering stage works on.

Resolution of the comparison. All results are averaged over five scenario seeds, which is enough to separate both placement strategies from the reference case but not enough to separate them from each other. Their accessibility differences are smaller than the spread across seeds. Reporting one as superior would require either more seeds or a scenario in which the two produce more distinct networks. Nor does the station count settle the question. Saturate-and-Prune ends wherever the prune loop is stopped, and the accessibility curve shows it passing through Clustering's operating point on the way down. What the data does support is a statement about effort, since Clustering obtains its network from a single simulation and Saturate-and-Prune from one per iteration. Station count also stands in for cost throughout. No cost model is involved, so a difference in stations translates into a difference in money only as long as the stations are comparable in power and connection effort. No published siting method was reimplemented and run against ours either. Areas, fleet sizes and reported metrics differ too widely between studies for the numbers to be comparable, and the aim here was a working pipeline and a procedure for locating capacity rather than a new optimiser to rank against existing ones.

One area, one fleet size. Every figure comes from the same square kilometer in Berlin with roughly 237 agents. The seeds redraw where those agents live and travel, not how many there are or what the road network looks like, so the reported spread quantifies population variation within one scenario and says nothing about transfer to other districts or fleet sizes. The claim that the datasets are generalizable to other urban areas remains untested. EV share, battery capacity and station power are pipeline parameters and were held fixed throughout, so how far the plateau moves with them is untested as well.

Charge-point sizing. The heuristic sizes a cluster by dividing its energy demand by the length of the observation window. Over a 24-hour horizon this averages away the peaks at which charging demand coincides, and consequently sizes every cluster at a single charge point regardless of how much energy it carries. A peak-based rule, applied to the busiest hour rather than to the daily mean, would give a more useful estimate.

Modelling detail. Two implementation choices deserve mention. Stations from OSM are point locations and get projected onto the nearest lane long enough to hold them, which introduces a small positional offset. That is harmless at the resolution of a siting question and would matter only in a study of access from individual parking spaces. Separately, the grid rating attached to a candidate site is a static estimate from the nominal capacity of the nearest bus, not a power flow along the feeder. A site rated excellent may still need reinforcement in operation. The power flow inside the simulation is a different model and does bound what each station may draw.

6. Conclusion

This paper presented a simulation-driven method for siting bidirectional charging stations from openly available data. Charging demand is derived from land use, resolved in space and time by microscopic traffic simulation, and turned into candidate sites by two competing placement strategies.

Across five scenario instances both strategies lift the fleet's end-of-day state of charge by roughly three percentage points and cut the share of undersupplied agents by six to eight. Neither beats the other. Adding stations accounts for about half of the gain and choosing them well for the other half. Accessibility also stops improving well before full coverage. For the area studied that ceiling sits near 75 stations. The practical value of the workflow is therefore less in ranking sites against each other than in locating the point where further construction stops paying off.

6.1 Future Work

Several directions follow from these results. Bidirectional operation stayed unused because discharge needs vehicles above 50% SoC while the vehicles at a station are there because they are low. Measuring what V2G is worth requires a fleet segment parked at a high charge when demand peaks, which points at evening home charging rather than public stations.

The sizing rule divides daily energy by the observation window and flattens the peaks a station has to serve. A rule built on the busiest hour would be more useful and needs no new simulation, only a different reading of the existing logs.

Finally, the plateau found here belongs to one square kilometer and one fleet size. Whether its position scales with density, area or EV share is the question that decides how far the method carries, and answering it means repeating the study across districts. While traffic is a key factor, direction of travel was not regarded in the placement heuristics. A more sophisticated algorithm could account for that characteristic of the traffic flow and increase the accuracy of the placement decisions.

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